



MODEL QUESTION PAPER

Course Code: CSE1012

Course Title: Problem solving using Python

Duration: 90 minutes

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Write an algorithm to read a multi-digit number. Swap the first and last digits of that number and display the updated number. **(10M)**

Example:

Input:4351 Output:1354

Q2. Draw a Flow chart to Find the sum of the given series. **(10M)**

$$\text{Result} = \sum_{k=m}^n (k^k + k!)$$

Example: m=2, n=5

$$\begin{aligned} \text{Result} &= (2^2+2!) + (3^3+3!) + (4^4+4!) + (5^5+5!) \\ &= 6+33+280+3245 = 3564 \end{aligned}$$

Q3. a. Find the octal and hexadecimal equivalent of $(25.75)_{10}$ **(5x2=10M)**

b. Do the number system conversion step by step process.
 $(256)_8 = ()_{16}$

c. Evaluate the expression step by step according to operator precedence and associativity and print the final output
 $4+3*2\&4|6$

d. Evaluate the expression step by step according to operator precedence and associativity and print the final output for x = 10 and y = 50
 $x ** 2 > 100 \text{ and } y < 100$

e. Write the output for the given code

```
a = 25
if a > 15:
    print("Hi")
if a <= 30:
    print("Hello")
else:
    print("Know Program")
```

Q4. Write a python program which prompts a user to enter the **Basic salary** to calculate the **Total salary** of an Employee considering the bonus and conveyance offered to him. Display the **Total salary** with two digits after the decimal point. **(10M)**

HRA = 20% of Basic

DA = 105 % of Basic

Conveyance = Rs1300

Bonus = Rs1000

Q5. Tirumala Tirupati Devasthanam's board offering a Divya Darshan tickets online booking. The fare and discount details are as follows: **(10M)**

Non Season Months

Jan - Mar	Adult	1000 INR	10% Discount
	Child	750 INR	10% Discount
Oct - Dec	Adult	750 INR	12% Discount
	Child	500 INR	12% Discount

Season Months

Apr - Jun	Adult	2000 INR	No Discount
	Child	1500 INR	No Discount
Jul - Sep	Adult	3000 INR	No Discount
	Child	2500 INR	No Discount

Write a python program to read month (in number), no. of adults and no. of children and find the total tickets booking price.

Input Output/ format:

Enter the month of Visit

5

Enter the number of Adult

2

Enter the number of child

1

Total Booking Price is Rs. 5500.00